Integrative Project ssignment

pi-sem2-2025-26 - version 2.0

April 14, 2026

Crystal Clear

bstract

This document comprises a detailed description of the Integrative Project Assignment for the practical work to be developed within ESOFT, ESTAT, MDISC, PPROG, and LAPR2 courses. The work consists of developing an IT solution that allows the capture, storage, and analysis of information related to the income, assets, interests, and connections of political actors. This document briefly introduces the business domain and sub-areas, the functional and non-functional requirements of the solution to be developed, the Integrative Project technical details, and the operating mode, i.e., the teamwork framework.

Table 1: Version register

Version	**Description**
1.0	First version
2.0	Sprint 2 US and Acceptance Criteria

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1 Integrative Project

In this project, students, organised into teams, will analyse, design, and develop a computer solution that allows the capture, storage, and analysis of information relating to the income, assets, interests, and connections of political actors, to promote transparency in the political system and bring those who govern closer to those they govern.

Following the good practices learned throughout the degree and in particular in Software Engineering (ESOFT), Programming Paradigms (PPROG), Statistics (ESTAT), Discrete Mathematics (MDISC), and Laboratory-Project II (LAPR2) courses, this project implies applying an iterative and incremental development process. Therefore, the agile SCRUM approach must be used to manage teamwork in each four-week SPRINT.

The software solution to be developed must be composed of a set of Java and Python applications that must meet the requirements. To increase the solution maintainability and follow good software development practices, the implementation must follow a Test-Driven Development (TDD) approach.

1.1 Team operation framework

Each team with LAPR2 students must nominate a Scrum Master for each Sprint (rotative role) and create a Scrum Board in the Organization *Departamento de Engenharia Informatica*, following these guidelines:

> Only one team member (the assigned scrum master) creates the Project Board with the title "sem2-pi-25-26-gnnn-board in which *nnn* must be replaced by the team number;
>
> The board privacy must be "private", and the "Organization Role" must be "no access";
>
> Access (writer) must be given to TP and PL teachers assigned to the team's class;
>
> The project board must include the following columns: **Backlog**, **To-Do**, **In Progress/Doing**, **Testing/To Review**, **Done**, **Logbook**.

All teams involved in the integrative project must create the project repository on GitHub in the Organization *Departamento de Engenharia Informatica*, following these guidelines:

> Only one team member creates the repository, with the title sem2-pi-25-26-gnnn-repo based on the provided **example**;

The repository access must be "private";

All TP, PL, and OT teachers from the involved class courses who supervise the project must be added to the repository with the "Reader" role.

2 Problem Statement: Description

In a democracy, the relationship between **transparency** and **political agents** (such as MP, ministers, and judges) is centered on the principle that those who hold public power must be subject to public scrutiny to prevent the abuse of that power for private gain.

The disclosure of their financial profiles, including **salaries, side incomes, and assets**, serves as a critical mechanism for maintaining democratic integrity through several key functions:

Preventing Conflicts of Interest and Illicit Enrichment

- **Asset Declarations**: These are "balance sheets" of an official's wealth (homes, stocks, debts) filed before, during, and after their term. By comparing these over time, oversight bodies and journalists can detect **illicit enrichment** (wealth that cannot be explained by the official salary).
- **Side Incomes**: Transparency regarding "side jobs" (consulting, board memberships) reveals potential **con icts of interest**. It allows voters to see if a politician's legislative actions are being influenced by private companies or lobby groups that pay them.

Enhancing Political Accountability and Trust

- **Trust-Building**: Research indicates that politicians who transparently disclose side incomes (even high ones) are viewed as **more trustworthy and electable** than those who refuse to disclose them based on "privacy concerns."
- **Public Oversight**: Because citizens e ectively pay the salaries of political agents through taxes, transparency ensures that these agents remain accountable to their "employers" (the public).

Strengthening the "Ballot Box" Mechanism Democracy relies on informed choice. Transparency provides the **information** necessary for

voters to make judgment calls at the polls. Without knowing a candidate's financial ties, voters cannot judge whether an official is serving the **public interest** or the interests of private donors.

International Standards and Best Practices

- **Global Frameworks**: The UN Convention against Corruption (UNCAC) requires countries to establish legal frameworks for public officials to declare assets and outside interests.
- **Digital Transparency**: Modern systems involve centralizing disclosures in **searchable, machine-readable portals** to allow for automated "red-flagging" of suspicious wealth increases.

2.1 Transparency Portal

The Portal for Transparency, the main deliverable of the current project, consists in an IT solution that supports the registration, evaluation, and analysis of Declarations of Interests (a.k.a. Asset Declaration) by political agent and the information resulting from their compilation and temporal analysis.

2.2 Declaration of Interest

Declarations of Interest may be **initial**, submitted by political agents whenever they begin a term of office; **regular**, submitted regularly (annually) while performing a political function; and **exceptional**, submitted whenever there is a significant change in the values mentioned in the declaration, when requested by the ethics committee, or for the purpose of correcting errors/omissions in previously submitted declarations.

Declarations of interest must be structured in sections and itens that include information regarding:

> professional positions, public, private, and social positions held (or previously held), in which institutions, the nature of these institutions, the functions performed, and the payment received;
>
> support and subsidies received, from which institutions and the nature of the institution;
>
> assets (real estate: urban and rural) with their estimated value;
>
> quotas, shares, and holdings in companies, with their respective market values.

2.3 Analysis Tools

The analysis tools are based on statistical methods and advanced algorithms, enabling them to highlight, in numerical and graphical form, relationships between political actors and organisations that may be difficult to discern through the individual analysis of declarations of interest. The dataset used by the analysis tools is compiled from all previously submitted declarations of interest that have been deemed valid. Furthermore, these tools allow the temporal dimension of the information to be extracted; when considering a single declaration of interests alone, this is restricted to a period of one year.

2.4 Platform Users and Stakeholders

This platform may be used by di erent users, namely:

Political Agent - someone who performs a political function, such as: deputy, minister, councillor, mayor, or parish council president;

Ordinary Citizen - a person who has an interest in scrutinising political agents and who can use the (platform) to report behaviour and processes that they consider to be lacking in transparency;

Journalist - a person who is registered with the Journalists' Union and wishes to access the Transparency Portal, on the basis that it is a fundamental tool for investigation and public scrutiny, supported by specific rights of access to administrative information;

Ethics Committee - normally responsible for: i) managing and supervising the Declarations of Interests of political agents; ii) verifying incompatibilities and impediments: situations where the professional or personal activity of a political agent may conflict with their mandate; iii) verifying the Code of Conduct: application and investigation of possible violations of the rules of conduct and civility;

System Administrator - person or team who manages the system and its access privileges.

3 Minimal Viable Product (MVP)

The purpose of this project is to develop a Minimal Viable Product (MVP) iteratively and incrementally; therefore, the work is divided into three Sprints:

Sprint 1 – Weeks 3 to 6 – from 9/March to 12/April

Sprint 2 – Weeks 7 to 10 – from 13/April to 17/May

Sprint 3 – Weeks 11 to 14 – from 18/May to 14/June

A description of the MVP is provided for each sprint. Teams must follow the user stories (US) provided and consider their interconnections and respective dependencies. At the end of each Sprint, each team must meet the specified requirements. Teams must add US to the backlog, size them appropriately, and distribute them among team members .

3.1 Sprint 1

3.1.1 Declaration of Interests and Analysis Tools

This Sprint aims at developing the following User Stories (US) (Requirement Engineering and Analysis):

US01 - As a future user, I want to request to be registered in the platform with the appropriate permissions for a specific role.

AC1: The role must be selected from a predefined list of available roles.

US02 - As an Administrator, I want to accept/reject registration requests on the platform.

US03 - As a Political Agent, I want to list Institutions (Companies, Political Parties, Foundations, Institutes, Associations).

AC1: The institutions should be grouped by type and then listed alphabetically by name.

US04 - As an Administrator, I want to register an Institution of a given type.

AC1: The institution type must be selected from a predefined list of available types.

US05 - As an Administrator, I want to register Functions/Positions.

US06 - As a Political Agent, I want to submit a declaration of interests including income, position/job/ post (public, private, or social), assets, and business participations.

US07 - As a Political Agent, I want to consult a previously submitted Declaration of Interests.

These will also be supervised in TP LAPR2 classes.

US08 - As a member of the Ethics Committee, I want to validate a submitted Declaration of Interests.

AC1: When correct, it is validated.

AC2: When incorrect, the section/item with inconsistencies must be commented.

US09 - As a member of the Ethics Committee, I want to consult the Integrated Situation of a Political Agent on a given date.

US10 - As a journalist, I want to analyse the evolution of a Political Agent's income over a given period of time (between two dates).

US11 - As a citizen/journalist, I want to consult the assets of a political agent on a specific date.

AC1: Sensitive data must be partially omitted based on the user's role.

US12 - As a citizen, I want to make a complaint about a political agent in a specific role and on a specific date.

For Sprint 1, regarding the Requirements Engineering activity:

teams with 5 or more students enrolled in ESOFT must elaborate all user stories;

teams with 4 students enrolled in ESOFT don't need to elaborate on the US05 and US07;

teams with 3 or fewer students enrolled in ESOFT don't need to elaborate on the US03, US04, US05, and US07.

3.2 Sprint 2

In addition to the user stories elaborated in Sprint 1, which will now follow the Software Engineering process and proceed to Design and Implementation phases, the following US are added in the current iteration (Sprint 2):

3.2.1 Descriptive Statistics

The statistical component allows the analysis of the political-declaration dataset by examining how declarations are distributed across types, roles, and institutions, and by computing key measures of central tendency and variability for total income and total assets, using each political agent's most

recent declaration. Additionally, the study of temporal trends in salaries, side incomes, and assets across all declaration types compares total income across roles using boxplots to assess medians, dispersion, and outliers, and investigates potential irregularities by identifying companies with the highest share values held by political agents and the agent (company pairs showing the largest increases in stock value over time). Together, these tasks apply core statistical methods to real political-transparency data.

Table 2 describes the variables and their corresponding types that characterize the **declaration dataset**, one row per declaration (initial, regular/annual, or exceptional).

Table 2: Declaration Dataset Variables

Column	**Type**	**Notes**
declaration id	string	Unique ID of the declaration
agent id	string	Identi ers
role	categorical	MP, Ministers, Judge, State Secretary, Advisor
institution	categorical	Parliament / Government / Courts
declaration type	categorical	Initial, Regular, Exceptional
declaration date	date	yyyy-mm-dd
gross salary	numeric	Annual gross salary reported
side income	numeric	It can be zero
assets in real estate	numeric	It can be zero
assets in vehicles	numeric	It can be zero
assets in stocks	numeric	It can be zero

The Table 3 describes the variables and their corresponding types that characterize the **holdings dataset**, one row per agent/company.

Table 3: Holdings Dataset Variables

Column	**Type**	**Notes**
agent id	string	Political actor identi ers
company NIF	numeric	Company associated
total value in stocks	numerical	
company percentage	numerical	% of the company
declaration date	date	yyyy-mm-dd

US13 - As a member of the Ethics Committee, I want to enhance my understanding of the dataset's structure; I want to get an exploratory

analysis of how declarations are distributed by type, by role, and by institution. My goal is to identify which roles and which institutions are responsible for submitting the highest number of declarations. To illustrate these patterns e ectively, use appropriate charts: one to present each declaration type proportion, and another to display the total number of declarations submitted by each role and by each institution.

US14 - As a member of the Ethics Committee, I want to get a comprehensive overview of the dataset, combining an analytical approach that examines central tendencies, dispersion, and overall distributional patterns. Appropriate measures of central and non-central location, as well as variability, must be computed for both total income and total assets. For each political actor, only the most recent declaration will be considered. To visually represent the distribution of these variables, suitable graphical techniques must be employed to illustrate both total income and total assets.

US15 - As a member of the Ethics Committee, I want to examine how a given political actor's total assets and net worth evolve over time. This involves producing three separate graphs: one for gross salary, one for side income, di erentiating among the various types of supplementary earnings, and one for assets, distinguishing the di erent categories of declared belongings. The charts should represent the evolution over time of the di erent analyzed parameters, taken from all declarations (initial, regular, and exceptional).

US16 - As a member of the Ethics Committee, I want to get a comparison of the total income (gross salary and side income) across roles; the analysis must use boxplots to reveal di erences in central tendency, variability, and the presence of outliers. The objective is to identify which role(s) exhibit the highest median incomes and which display the greatest dispersion, while also listing and interpreting any detected outliers.

US17 - As a member of the Ethics Committee, aiming to conduct a more rigorous analysis of the relationship between political actors and companies, and to identify any potentially suspicious connections, the objective is to generate a chart displaying the ten companies with the highest values of shares held by political actors, along with their respective total amounts. For each political actor, only the most recent declaration will be considered.

> US18 - As a member of the Ethics Committee, I want to identify the ten political-actor/company pairs that show the largest increase in the value of the actor's stocks in the company, from their initial declaration to their most recent declaration. For the pairs identified, I want to see their evolution over time.

3.2.2 Heterogeneous and multi-relational networks

We want to improve the Transparency Portal, the main deliverable of the current project, by creating a visualisation of a relations map.

Heterogeneous and multi-relational networks are crucial structures in modern data analysis, modeling complex systems in which relationships are not uniform.

> A *heterogeneous network* is a graph that combines di erent types of vertices (entities, *e.g.,* person, organization, position/job, asset).
>
> A *multi-relational network* is a graph in which pairs of vertices can be connected by multiple distinct types of edges (relationships, *e.g.,* (person, relativeOf, person), (person, occupies, job), (person, memberOf, organization), (job, inOrganization, organization), (organization, regulates, organization)).

Edges can have, or not, directions and/or weights (intensity of the relationship). Both entities and relationships have life cycles: they emerge, evolve, and eventually terminate over time.

Notes for creating these networks:

(i) User permission restrictions will not be considered.

(ii) Global Acceptance Criteria: adjacency matrices will be used to store the information of the graph to be visualised, as well as to perform calculations/analyses; .csv files with information about entities and relationships will populate these matrices.

The following US are designed to support the construction of a heterogeneous and multi-relational graph at a given instant in time. Accordingly, the model is based on discrete time snapshots, where each snapshot represents the state of a dynamic graph, forming a temporal sequence of network states.

> US19 - As a Product Owner, I want to extract the entities with the information from an imported .csv file.

AC1: In the graph drawing, entities of the same type must have the same symbol, which must be di erent from the symbols of the other entity types.

AC2: Entities have di erent types, where *type* is in {person (politician, family member, businessman, advisor, . . .), organization (company, public institution, political party, NGO, foundation, . . .), position (public, business, political party, . . .), asset (company, property, fund, . . .)}.

The format of an entity is the following:

- person: [id, type, startDate, endDate, name, birthDate, nationality];
- organization: [id, type, startDate, endDate, name, organizationType, country];
- position: [id, type, startDate, endDate, positionTitle, positionType, organizationId];
- asset: [id, type, startDate, endDate, assetType, country, estimatedValue].

US20 - As a Product Owner, I want to draw the relationships between entities with the information from an imported .csv file.

AC1: In the graph, the di erent types of relationships should be represented with di erent colors or lines.

AC2: All relationships have a common structure:
[id, type, entity1,entity2, weight, startDate, endDate].

AC3: Types of relationships to consider[2]:

- Person - Person relationships (disjoint relations): relativeOf, friendOf, associatedWith, appointedBy;
- Person - Position relationship: holdsPosition;

[2]Ethical and legal considerations: In a network analysis system involving political agents and sensitive relationships, it is crucial to ensure a clear distinction between veri ed facts and allegations or inferences, avoiding presenting uncon rmed information as established truth. Special care must be taken regarding defamation risks, ensuring that all published information is supported by reliable sources and properly contextualized. Furthermore, data processing must strictly comply with the General Data Protection Regulation (GDPR), especially regarding the lawfulness of processing, data minimisation, transparency, and the rights of data subjects, particularly when personal or potentially sensitive data is involved.

- Position - Organization relationship: inOrganization;
- Organization - Organization relationships (disjoint relations): controls, partnerOf, supervises;
- Person - Organization relationship: influences;
- Person - Political Party (Organization) relationships (disjoint relations): memberOf, supports, opposes;
- Person - Asset relationships (disjoint relations): ownerOf, partnerOf.

US21 - For a given temporal snapshot, as a Product Owner, I aim to represent the graph using two types of adjacency matrices:

- (i) for each relation r, the information is stored in a matrix of dimension $m \quad n$;
- (ii) a global matrix of dimension $N \quad N$, where N is the total number of entities. The first provides better interpretability (avoiding invalid entries, preserving directed or undirected edges, and facilitating some calculations), while the latter o ers a more compact and unified representation of the overall network structure.

Instead of binary, the entries in each matrix have a weight that represents the strength of the relationship, the percentage of participation, among other aspects.

The following User Stories are designed to enable users and stakeholders to interact with the platform.

US22 - As a member of the Ethics Committee, I want to find out if someone was appointed by a relative, friend, or associate.

AC1: All pairs of individuals involved in direct nepotism must be listed.

US23 - As a member of the Ethics Committee, I want to detect potential (indirect) conflicts of interest by asking specific questions, such as:

- (Nepotism) "Does a certain person have relatives who hold prominent positions?", or "Does a certain person have relatives who hold prominent positions in an organization?"
- (Conflict of interest) "Does a person influence a company through a public position?"

AC1: The question must be selected from a predefined list of available questions (at least 3).

AC2: If the answer is *yes*, the first and last entities in the chain must be revealed.

3.2.3 Integration Aspects

US24 - As Product Owner, I want to extract information from the Declarations of Interest and export it to a .csv file, according to the data set described in Table 2;

US25 - As Product Owner, I want to draw information from Declarations of Interest and export it to a .csv file according to the data set described in Table 3.

US26 - As Product Owner, I want to interact with the heterogeneous and multi-relational graph using hyperlinks.

AC1: The hyperlinks for entities and relations should depict their details.

3.3 Non-functional requirements

This section describes some non-functional requirements that must be considered when implementing the project.

Validation of business rules must be respected when recording and updating data.

The class structure must be designed to allow easy maintenance and the addition of new features following the best Object-Oriented (OO) practices.

The application must be developed in Java language.

All those who wish to use the application must be authenticated with a password of seven alphanumeric characters, including three capital letters and two digits.

The information provided varies depending on the type of user.

The application documentation must be in English.

During system development, the team must: (i) adopt best practices for identifying requirements and for OO software analysis and design; (ii) adopt recognized coding standards (e.g., CamelCase); (iii) use Javadoc to generate useful documentation for Java code.

The development team must implement unit tests for all methods, except for the methods that implement Input/Output operations. Unit tests should be implemented using the JUnit 5 framework. The JaCoCo plugin should be used to generate the coverage report.

All the images/figures produced during the software development process should be recorded in SVG format.